

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 17 September 2026, 8:09 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:18 AM**Time taken** 9 mins 6 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("");
                printf("");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      int count = 0;
6
7      // Read the positive integer n
8      if (scanf("%d", &n) != 1) {
9          return 0;
10     }
11
12     count++; // 1 for the if(n==1) condition check
13     if (n == 1) {
14         count++; // 1 for the printf("") operation inside the block
15     }
16     else {
17         count++; // 1 for the outer loop initialization (int i = 1)
18
19         for (int i = 1; i <= n; i++) {
20             count++; // 1 for outer loop condition check (i <= n)
21             count++; // 1 for inner loop initialization (int j = 1)
22
23             for (int j = 1; j <= n; j++) {
24                 count++; // 1 for inner loop condition check (j <= n)
25                 count++; // 1 for the print block execution
26                 break;
27             }
28         }
```

```
29         count++; // 1 for outer loop increment (i++)
30     }
31 }
32
33 // Print the final count value as required
34 printf("%d", count);
35
36 return 0;
37 }
38
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)